# Category Theory for Mathematical Physics — Exercise Sheet 2

*Universal properties, adjunctions, semisimple categories*

**Variant: Solutions**

Conventions as in Lectures 1 and 2: in $g \circ f$ the morphism $f$ acts first; $\mathrm{Rep}(G)$ has the finite-dimensional complex representations; the simple objects of $\mathrm{Rep}(S_3)$ are $\mathbf{1}$, sgn and std (Example 2.15); $\mathbb{Z}$ and $\mathbb{Z}_n$ are written additively; $\zeta_n = e^{2\pi i/n}$. Characters and their inner product $\langle \chi, \chi' \rangle = \frac{1}{|G|} \sum_g \overline{\chi(g)} \chi'(g)$ are assumed from *Symmetries*.

**Exercise 1 (Universal properties in the divisibility category).** Let Div have objects $0, 1, 2, \ldots$ and exactly one morphism $m \to n$ when $m$ divides $n$ (so every $m$ divides 0, and 0 divides only 0), and none otherwise.

(i) Show that Div is a category. Find its initial and its terminal object.

(ii) Show that the product of $m$ and $n$ is $\gcd(m, n)$ and the coproduct is $\mathrm{lcm}(m, n)$, with $\gcd(m, 0) = m$ and $\mathrm{lcm}(m, 0) = 0$.

(iii) Lemma 2.2 says a terminal object is unique up to a unique isomorphism. What are the isomorphisms of Div, and what does the lemma say here? In Set, show that the terminal objects are the one-point sets and the only initial object is $\emptyset$; in Vect, show that the zero space is both.

**Solution.** (i) $m \mid m$ gives the identities, and $m \mid n$, $n \mid p$ give $m \mid p$, the composite. With at most one morphism between two objects, any two morphisms with the same source and target agree, so associativity and the unit laws hold. Since 1 divides every $n$, there is exactly one morphism $1 \to n$: 1 is initial. Since every $n$ divides 0, there is exactly one morphism $n \to 0$: 0 is terminal.

(ii) A product of $m$ and $n$ is an object $P$ with morphisms $P \to m$, $P \to n$, that is $P \mid m$ and $P \mid n$, such that every $Z$ with $Z \mid m$, $Z \mid n$ has a morphism $Z \to P$, $Z \mid P$ (it is then unique, and the triangles commute because every diagram does). So $P$ is a common divisor that every common divisor divides. For $m, n \geq 1$, $\gcd(m, n)$ is such: every common divisor divides it, by Bézout's identity $\gcd(m, n) = am + bn$. If $n = 0$, the common divisors of $m$ and 0 are the divisors of $m$, and $m$ is the one they all divide: $\gcd(m, 0) = m$ (and $\gcd(0, 0) = 0$). Dually, a coproduct $C$ has $m \mid C$, $n \mid C$, and $C \mid Z$ for every common multiple $Z$. For $m, n \geq 1$, $\mathrm{lcm}(m, n)$ divides every common multiple. If $n = 0$, the only common multiple of $m$ and 0 is 0, so $C = 0$.

(iii) A morphism $m \to n$ with an inverse means $m \mid n$ and $n \mid m$, so $m = n$: the only isomorphisms are the identities, and the unique isomorphism of the lemma is an identity. The terminal object is literally unique, and so are products and coproducts. In Set, a set $T$ is terminal exactly when it has one point: every set has exactly one function to a one-point set, while if $T = \emptyset$ there is no function from $\{*\}$ to $T$, and if $T$ has at least two points there are at least two. Any two one-point sets are isomorphic, by exactly one bijection. A set is initial exactly when it is $\emptyset$: there is one (empty) function from $\emptyset$ to each set, and a nonempty set has no function to $\emptyset$. In Vect the zero space is both initial and terminal: a linear map $0 \to V$ must send 0 to 0, and the only map $V \to 0$ is zero.

**Exercise 2 (Yoneda, in full).** (i) Prove Theorem 2.6: for a functor $\Phi \colon C \to \mathrm{Set}$ and an object $X$, the natural transformations $\xi \colon \mathrm{Hom}(X, -) \Rightarrow \Phi$ correspond to the elements of $\Phi(X)$ by $\xi \mapsto \xi_X(\mathrm{id}_X)$. (Construct the inverse: for $a \in \Phi(X)$, put $\xi^a_Y(f) = \Phi(f)(a)$.)

(ii) Let $U\colon \mathrm{Vect}_{\mathrm{fd}} \to \mathrm{Set}$ forget the linear structure. Show that $U$ is represented by $\mathbb{C}$, and deduce that the natural transformations $U \Rightarrow U$ are exactly the families $v \mapsto \lambda v$, one $\lambda \in \mathbb{C}$ for all spaces at once. Show directly that $v \mapsto v + c_V$, with a nonzero vector $c_V$ chosen in each nonzero $V$, is not natural.

(iii) Show that the natural transformations $\mathrm{Hom}(V,-) \Rightarrow \mathrm{Hom}(W,-)$ on $\mathrm{Vect}_{\mathrm{fd}}$ are exactly the maps $f \mapsto f \circ \psi$, one for each linear $\psi\colon W \to V$.

**Solution.** (i) For $a \in \Phi(X)$ the maps $\xi^a_Y(f) = \Phi(f)(a)$ are natural: for $g\colon Y \to Y'$, $\Phi(g)(\xi^a_Y(f)) = \Phi(g)\Phi(f)(a) = \Phi(g \circ f)(a) = \xi^a_{Y'}(g \circ f)$, which is the naturality square applied to $f$. And $\xi^a_X(\mathrm{id}_X) = \Phi(\mathrm{id}_X)(a) = a$. Conversely, for a natural $\xi$ put $a = \xi_X(\mathrm{id}_X)$. Naturality at $f\colon X \to Y$, applied to $\mathrm{id}_X$, gives

$$\xi_Y(f) = \xi_Y(f \circ \mathrm{id}_X) = \Phi(f)\big(\xi_X(\mathrm{id}_X)\big) = \Phi(f)(a) = \xi^a_Y(f),$$

so $\xi = \xi^a$. The two assignments are inverse to each other.

(ii) The maps $\beta_V\colon \mathrm{Hom}(\mathbb{C}, V) \to V$, $f \mapsto f(1)$, are bijections (a linear map on $\mathbb{C}$ is fixed by the image of 1), and natural: for $g\colon V \to W$, $\beta_W(g \circ f) = g(f(1)) = g(\beta_V(f))$. So $U \cong \mathrm{Hom}(\mathbb{C}, -)$, and natural transformations $U \Rightarrow U$ correspond to those $\mathrm{Hom}(\mathbb{C},-) \Rightarrow U$, which by (i) correspond to the elements $\lambda$ of $U(\mathbb{C}) = \mathbb{C}$. The one for $\lambda$ is $\xi^\lambda_V(f) = U(f)(\lambda) = f(\lambda) = \lambda f(1)$, which through $\beta$ is $v \mapsto \lambda v$. Directly: a natural family of functions $\xi_V\colon V \to V$ must satisfy $f(\xi_V(v)) = \xi_W(f(v))$ for every linear $f$; with $f = 0\colon V \to W$ and $W \neq 0$ this gives $0 = \xi_W(0)$. For $\xi_W(w) = w + c_W$ that reads $0 = c_W \neq 0$, a contradiction.

(iii) By (i) with $C = \mathrm{Vect}_{\mathrm{fd}}$, $X = V$ and $\Phi = \mathrm{Hom}(W,-)$, the natural transformations correspond to $\psi \in \Phi(V) = \mathrm{Hom}(W,V)$, and the one for $\psi$ is $\xi^\psi_Y(f) = \mathrm{Hom}(W,f)(\psi) = f \circ \psi$. So $V \mapsto \mathrm{Hom}(V,-)$ reverses arrows and, on morphisms, is a bijection: the representing object is recovered from the functor, as in Proposition 2.7.

**Exercise 3 (The free vector space).** For a set $X$ let $\mathbb{C}[X]$ be the space of functions $X \to \mathbb{C}$ that vanish outside a finite set, with basis $e_x$ ($x \in X$); for a function $\alpha\colon X \to X'$ let $F(\alpha)\colon \mathbb{C}[X] \to \mathbb{C}[X']$, $e_x \mapsto e_{\alpha(x)}$. Let $U\colon \mathrm{Vect} \to \mathrm{Set}$ forget the linear structure.

(i) Show that $F\colon \mathrm{Set} \to \mathrm{Vect}$ is a functor.

(ii) Show that $F \dashv U$: $f \mapsto (x \mapsto f(e_x))$ is a bijection $\mathrm{Hom}_{\mathrm{Vect}}(\mathbb{C}[X], V) \to \mathrm{Hom}_{\mathrm{Set}}(X, U(V))$, natural in $X$ and $V$. Find the unit $\eta_X$ and the counit $\varepsilon_V$, check the two triangle identities $\varepsilon_{F(X)} \circ F(\eta_X) = \mathrm{id}_{F(X)}$ and $U(\varepsilon_V) \circ \eta_{U(V)} = \mathrm{id}_{U(V)}$, and show that the counit $\mathbb{C}[U(\mathbb{C})] \to \mathbb{C}$ is not injective.

(iii) For $|X| = 3$ and $V = \mathbb{C}^2$, what is $\dim \mathrm{Hom}_{\mathrm{Vect}}(\mathbb{C}[X], V)$?

**Solution.** (i) $F(\beta \circ \alpha)e_x = e_{\beta(\alpha(x))} = F(\beta)F(\alpha)e_x$ and $F(\mathrm{id}_X)e_x = e_x$; both sides are linear, so they agree on $\mathbb{C}[X]$.

(ii) Write $\Theta(f)(x) = f(e_x)$. A linear map out of $\mathbb{C}[X]$ is determined by its values on the basis $(e_x)$, and every function $\varphi\colon X \to U(V)$ extends to the linear map $e_x \mapsto \varphi(x)$; so $\Theta$ is a bijection. Natural in $X$: for $\alpha\colon X' \to X$, $\Theta(f \circ F(\alpha))(x') = f(e_{\alpha(x')}) = (\Theta(f) \circ \alpha)(x')$. Natural in $V$: for linear $b\colon V \to V'$, $\Theta(b \circ f) = U(b) \circ \Theta(f)$. The unit is $\eta_X = \Theta(\mathrm{id}_{\mathbb{C}[X]})\colon X \to U(\mathbb{C}[X])$, $x \mapsto e_x$. The counit is $\varepsilon_V = \Theta^{-1}(\mathrm{id}_{U(V)})\colon \mathbb{C}[U(V)] \to V$, the linear map with $e_v \mapsto v$. Triangle identities: $\varepsilon_{F(X)} \circ F(\eta_X)$ sends $e_x$ to $e_{e_x}$ (the basis vector of $\mathbb{C}[U(\mathbb{C}[X])]$ labelled by the vector $e_x$) and then to $e_x$, so it is $\mathrm{id}_{F(X)}$; and $U(\varepsilon_V) \circ \eta_{U(V)}$ sends $v$ to $e_v$ and then to $v$, so it is $\mathrm{id}_{U(V)}$. In $\mathbb{C}[U(\mathbb{C})]$ the vectors $e_1$ and $e_2$ are independent basis vectors, and $\varepsilon_{\mathbb{C}}(e_2 - 2e_1) = 2 - 2 \cdot 1 = 0$: the counit is not injective. $\mathbb{C}[U(V)]$ has one basis vector for every vector of $V$, and forgets that $e_{v+w}$ and $e_v + e_w$ should agree.

(iii) $\Theta$ identifies $\mathrm{Hom}_{\mathrm{Vect}}(\mathbb{C}[X], \mathbb{C}^2)$ with all functions $X \to \mathbb{C}^2$, a space of dimension $|X| \cdot 2 = 6$ (and $\Theta$ is linear).

**Exercise 4 (The Hom-dimension table of $\mathrm{Rep}(S_3)$).** In $\mathrm{Rep}(S_3)$ take the simple objects $\mathbf{1}$, sgn, std and the permutation representations $P = \mathbb{C}^3$ (on $\{1,2,3\}$), $Y = \mathbb{C}[S_3/A_3]$ (on the two cosets of $A_3$, swapped by the odd permutations) and $R = \mathbb{C}[S_3]$ (the regular representation).

(i) Find the multiplicities of $\mathbf{1}$, sgn, std in $P$, $Y$ and $R$.

(ii) Fill in the $6 \times 6$ table of $\dim \mathrm{Hom}(X, X')$ for $X, X'$ among $\mathbf{1}$, sgn, std, $P, Y, R$.

(iii) For which of the six is $\mathrm{End}(X)$ commutative?

**Solution.** (i) On the classes of 1, the transpositions and the 3-cycles (sizes $1, 3, 2$), the characters are $\chi_{\mathbf{1}} = (1,1,1)$, $\chi_{\mathrm{sgn}} = (1,-1,1)$, $\chi_{\mathrm{std}} = (2,0,-1)$, and the permutation characters count fixed points: $\chi_P = (3,1,0)$, $\chi_Y = (2,0,2)$ (a transposition swaps the cosets, a 3-cycle fixes both), $\chi_R = (6,0,0)$. The inner products give

$$m(P) = \left(\tfrac{3+3}{6}, \tfrac{3-3}{6}, \tfrac{6}{6}\right) = (1,0,1),$$
$$m(Y) = \left(\tfrac{2+4}{6}, \tfrac{2+4}{6}, \tfrac{4-4}{6}\right) = (1,1,0),$$
$$m(R) = \left(\tfrac{6}{6}, \tfrac{6}{6}, \tfrac{12}{6}\right) = (1,1,2).$$

(ii) With $m(\mathbf{1}) = (1,0,0)$, $m(\mathrm{sgn}) = (0,1,0)$, $m(\mathrm{std}) = (0,0,1)$ and (i), each entry is a dot product of multiplicity vectors, by Theorem 2.13:

| | $\mathbf{1}$ | sgn | std | $P$ | $Y$ | $R$ |
|---|---|---|---|---|---|---|
| $\mathbf{1}$ | 1 | 0 | 0 | 1 | 1 | 1 |
| sgn | 0 | 1 | 0 | 0 | 1 | 1 |
| std | 0 | 0 | 1 | 1 | 0 | 2 |
| $P$ | 1 | 0 | 1 | 2 | 1 | 3 |
| $Y$ | 1 | 1 | 0 | 1 | 2 | 2 |
| $R$ | 1 | 1 | 2 | 3 | 2 | 6 |

The table is symmetric, since the formula is. The entry $\dim \mathrm{Hom}(P,R) = 3$ is the one computed in Example 2.15, and $\dim \mathrm{End}(R) = 6 = |S_3|$.

(iii) By Theorem 2.13, $\mathrm{End}(X) \cong \prod_i M_{m_i}(\mathbb{C})$, which is commutative exactly when every $m_i \le 1$. That holds for $\mathbf{1}$, sgn, std, $P$ and $Y$, and fails for $R$, where std has multiplicity 2: $\mathrm{End}(R) \cong \mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$, as in Example 2.14.

**Exercise 5 (Semisimple, or not).** (i) Show that $\mathrm{Rep}(\mathbb{Z}_n)$ is semisimple with $n$ simple objects, and that $\mathrm{Rep}(\mathbb{Z}_2)$ and $\mathrm{Rep}(\mathbb{Z}_3)$ are not equivalent as linear categories. You may use that an equivalence is full, faithful and essentially surjective (Leinster Prop. 1.3.18, the converse of Proposition 1.17).

(ii) An invertible $2 \times 2$ matrix $A$ gives a representation $\rho_A(k) = A^k$ of $\mathbb{Z}$ on $\mathbb{C}^2$. Show that $\rho_A$ is a direct sum of two one-dimensional representations exactly when $A$ is diagonalisable. Compute $\mathrm{End}(\rho_A)$, the matrices commuting with $A$, for $A_1 = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, and their dimensions.

(iii) Show that $\mathrm{End}(\rho_{A_2})$ is not isomorphic, as an algebra, to any product of matrix algebras $\prod_i M_{m_i}(\mathbb{C})$. Conclude that no semisimple linear category contains an object with this endomorphism algebra.

**Solution.** (i) $\mathbb{Z}_n$ is finite, so by Maschke and Schur (as for $\mathrm{Rep}(G)$ in Lecture 2) $\mathrm{Rep}(\mathbb{Z}_n)$ is semisimple, with as many simple objects as conjugacy classes: $n$, the group being abelian. They are the characters $k \mapsto \zeta_n^{jk}$, $j = 0, \dots, n-1$. In a semisimple category, Theorem 2.13 gives $\mathrm{End}(X) \cong \prod_i M_{m_i}(\mathbb{C})$, so $\mathrm{End}(X) \cong \mathbb{C}$ exactly when one multiplicity is 1 and the others 0, that is, when $X$ is isomorphic to a simple object. An equivalence $F$ is full and faithful,

so $\mathrm{End}(F(X)) \cong \mathrm{End}(X)$ as algebras, and it keeps isomorphic objects isomorphic and is essentially surjective; so it induces a bijection between the isomorphism classes of objects with End $\cong \mathbb{C}$. These number 2 in $\mathrm{Rep}(\mathbb{Z}_2)$ and 3 in $\mathrm{Rep}(\mathbb{Z}_3)$, so no equivalence exists.

(ii) A one-dimensional subrepresentation is a line $L$ with $\rho_A(k)L = L$ for all $k$, that is, $AL = L$: an eigenline of $A$. If $A$ has two independent eigenvectors, their lines are subrepresentations and $\mathbb{C}^2$ is their direct sum. Conversely, if $\mathbb{C}^2 = L_1 \oplus L_2$ with each $L_j$ invariant, a nonzero vector from each is an eigenvector, and they form an eigenbasis. An intertwiner $\rho_A \to \rho_A$ is a matrix $T$ with $TA^k = A^kT$ for all $k$, that is, $TA = AT$. For $A_1$, with distinct eigenvalues 2 and 3, the commuting matrices are those diagonal in an eigenbasis: $\mathrm{End}(\rho_{A_1}) \cong \mathbb{C} \times \mathbb{C}$, of dimension 2. Explicitly, $T = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ commutes with $A_1$ exactly when $c = 0$ and $d = a + b$. For $A_2 = 2I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $T$ commutes with $A_2$ exactly when it commutes with $N$, which forces $T = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} = aI + bN$: again dimension 2. $A_2$ has the single eigenline $\mathbb{C}e_1$, so $\rho_{A_2}$ is not a sum of two lines (Example 2.16).

(iii) $\mathrm{End}(\rho_{A_2}) = \{aI + bN\}$ is commutative and contains $N \neq 0$ with $N^2 = 0$. A product $\prod_i M_{m_i}(\mathbb{C})$ is commutative only if every $m_i \leq 1$, and then it is $\mathbb{C}^k$, in which $x^2 = 0$ forces $x = 0$. So no such isomorphism exists. In a semisimple category every object has End $\cong \prod_i M_{m_i}(\mathbb{C})$ by Theorem 2.13, so no object there has this endomorphism algebra. The dimension of End alone does not tell $\rho_{A_1}$ from $\rho_{A_2}$; the algebra does.

**Exercise 6 ⋆.** *Frobenius reciprocity, built by hand.* Let $G = S_3$ and $H = \{1, (1\,2)\} \cong S_2$. For a representation $W$ of $H$, let $\mathrm{Ind}\, W = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ be the space $\mathbb{C}[G] \otimes W$ divided by the span of all $gh \otimes w - g \otimes hw$ ($g \in G$, $h \in H$, $w \in W$), with $G$ acting by $g'(g \otimes w) = g'g \otimes w$.

(i) Let $g_1, g_2, g_3$ be representatives of the cosets $gH$. Show that $(w_1, w_2, w_3) \mapsto \sum_i g_i \otimes w_i$ is a linear isomorphism $W^3 \to \mathrm{Ind}\, W$, so $\dim \mathrm{Ind}\, W = 3 \dim W$.

(ii) On an $H$-intertwiner $a\colon W' \to W$ let $\mathrm{Ind}(a)(g \otimes w') = g \otimes a(w')$. Show that this is well defined, and that $f \mapsto \big(w \mapsto f(1 \otimes w)\big)$ is a bijection $\mathrm{Hom}_G(\mathrm{Ind}\, W, V) \to \mathrm{Hom}_H(W, \mathrm{Res}\, V)$, natural in $W$ and in $V$: so Ind $\dashv$ Res.

(iii) Use (ii) to decompose $\mathrm{Ind}\, \mathbf{1}_H$ and $\mathrm{Ind}\, \mathrm{sgn}_H$ into $\mathbf{1}$, sgn, std.

**Solution.** (i) Every $g \in G$ is $g_i h$ for exactly one $i$ and one $h \in H$. In the quotient, $g \otimes w = g_i h \otimes w = g_i \otimes hw$, so the map $(w_i) \mapsto \sum_i g_i \otimes w_i$ is onto. Define $\Psi\colon \mathbb{C}[G] \otimes W \to W^3$ by sending $g \otimes w$, with $g = g_i h$, to $hw$ in slot $i$ (and extending linearly). A relation $gh' \otimes w - g \otimes h'w$ with $g = g_i h$ goes to $hh'w$ minus $h(h'w)$ in slot $i$: zero. So $\Psi$ descends to $\mathrm{Ind}\, W$, and it inverts the map: $\Psi(\sum_i g_i \otimes w_i) = (w_i)$, while $\sum_i g_i \otimes \Psi(g \otimes w)_i = g_i \otimes hw = g \otimes w$. Hence $\dim \mathrm{Ind}\, W = 3 \dim W$.

(ii) For a $G$-intertwiner $f$, $\varphi(w) = f(1 \otimes w)$ is an $H$-intertwiner: $\varphi(hw) = f(1 \otimes hw) = f(h \otimes w) = \rho_V(h) f(1 \otimes w) = \rho_V(h)\varphi(w)$. Conversely, for an $H$-intertwiner $\varphi$ put $f_\varphi(g \otimes w) = \rho_V(g)\varphi(w)$. It respects the relations, $f_\varphi(gh \otimes w) = \rho_V(g)\rho_V(h)\varphi(w) = \rho_V(g)\varphi(hw) = f_\varphi(g \otimes hw)$, so it is defined on $\mathrm{Ind}\, W$, and it is a $G$-intertwiner, $f_\varphi(g'g \otimes w) = \rho_V(g') f_\varphi(g \otimes w)$. The two maps are inverse: $f_\varphi(1 \otimes w) = \varphi(w)$, and $f(g \otimes w) = f(g \cdot (1 \otimes w)) = \rho_V(g) f(1 \otimes w)$. $\mathrm{Ind}(a)$ is well defined: it sends a relation $gh \otimes w' - g \otimes hw'$ to $gh \otimes a(w') - g \otimes a(hw') = gh \otimes a(w') - g \otimes h\, a(w')$, again a relation, since $a$ is an $H$-intertwiner; and it commutes with the action of $G$. Naturality: for a $G$-intertwiner $b\colon V \to V'$, both routes send $f$ to $w \mapsto b(f(1 \otimes w))$; for an $H$-intertwiner $a\colon W' \to W$, both routes send $f$ to $w' \mapsto f(1 \otimes a(w'))$.

(iii) $\mathrm{Rep}(S_3)$ is semisimple, so the multiplicity of a simple $V$ in $\mathrm{Ind}\, W$ is $\dim \mathrm{Hom}_G(\mathrm{Ind}\, W, V)$ (the formula of Theorem 2.13 with $V$ simple), which by (ii) is $\dim \mathrm{Hom}_H(W, \mathrm{Res}\, V)$. Restricted to $H$, $\mathbf{1}$ is $\mathbf{1}_H$, sgn is $\mathrm{sgn}_H$ (the transposition is odd), and std, of character 2 at 1 and

0 at $(1\,2)$, is $\mathbf{1}_H \oplus \mathrm{sgn}_H$. So $\mathrm{Ind}\,\mathbf{1}_H$ contains $\mathbf{1}$ once, sgn never and std once: $\mathrm{Ind}\,\mathbf{1}_H = \mathbf{1} \oplus \mathrm{std}$, of dimension 3 as (i) requires (it is the permutation representation on $G/H \cong \{1,2,3\}$). And $\mathrm{Ind}\,\mathrm{sgn}_H$ contains $\mathbf{1}$ never, sgn once and std once: $\mathrm{sgn} \oplus \mathrm{std}$.